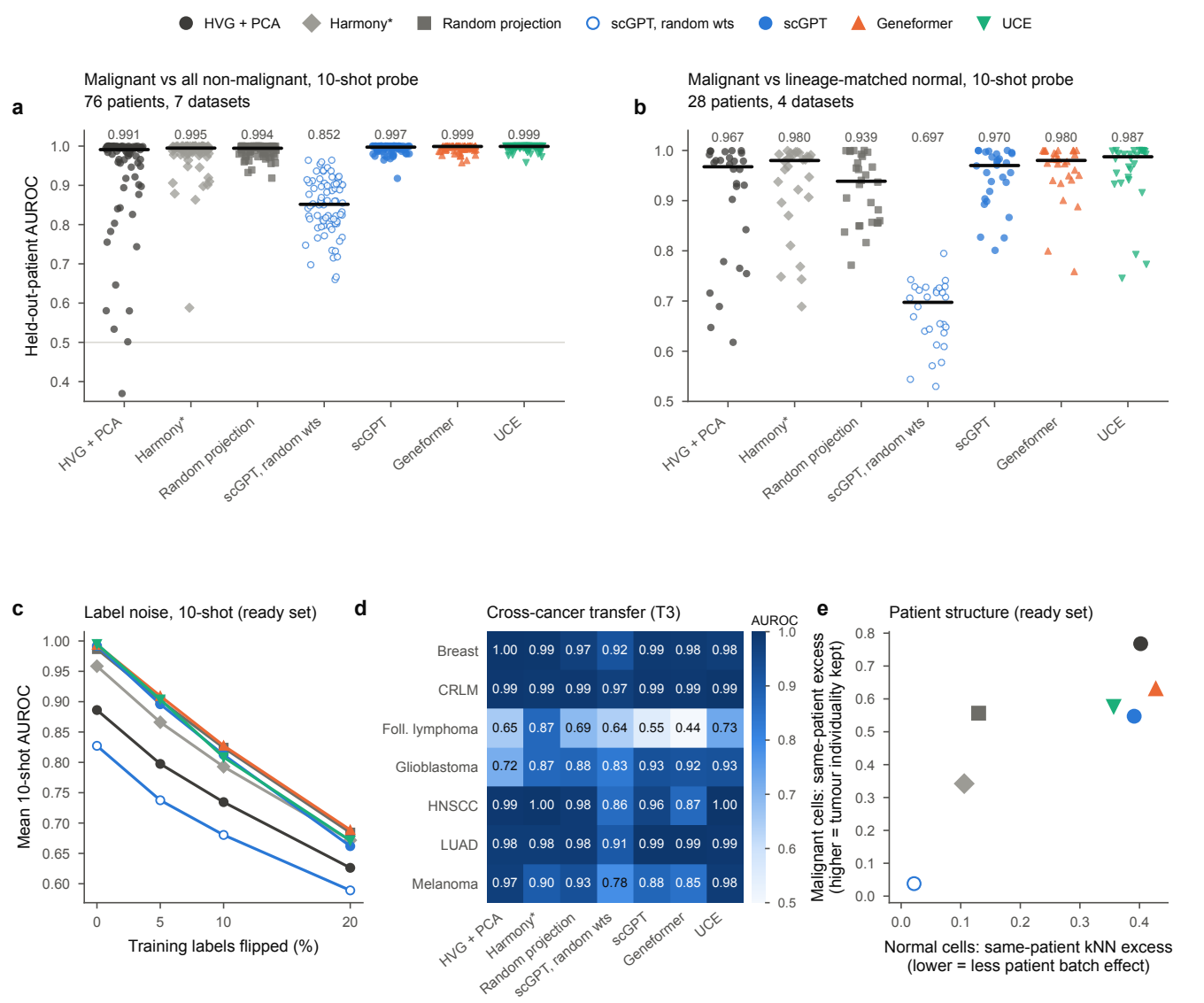

HVG + PCA
Harmony*
Random projection
scGPT, random wts
scGPT
Geneformer
UCE
a
Malignant vs all non-malignant, 10-shot probe
76 patients, 7 datasets
Held-out-patient AUROC
0.991
0.995
0.994
0.852
0.997
0.999
0.999
b
Malignant vs lineage-matched normal, 10-shot probe
28 patients, 4 datasets
0.967
0.980
0.939
0.697
0.970
0.980
0.987
c
Label noise, 10-shot (ready set)
Mean 10-shot AUROC
Training labels flipped (%)
d
Cross-cancer transfer (T3)
AUROC
Breast 1.00 0.99 0.97 0.92 0.99 0.98 0.98
CRLM 0.99 0.99 0.99 0.97 0.99 0.99 0.99
Foll. lymphoma 0.65 0.87 0.69 0.64 0.55 0.44 0.73
Glioblastoma 0.72 0.87 0.88 0.83 0.93 0.92 0.93
HNSCC 0.99 1.00 0.98 0.86 0.96 0.87 1.00
LUAD 0.98 0.98 0.98 0.91 0.99 0.99 0.99
Melanoma 0.97 0.90 0.93 0.78 0.88 0.85 0.98
e
Patient structure (ready set)
Malignant cells: same-patient excess
(higher = tumour individuality kept)
Normal cells: same-patient kNN excess
(lower = less patient batch effect)